Electronic Devices and Circuit Theory

Today's Topics:

- Semiconductors
- Diodes

Important!

- **The first lab will be at 5th Week!**
  ==Next Week
- **The second lab will be at 7th Week!**

You need to attend labs!

However, you are allowed to miss up to two lab sessions.

If you miss more than two lab sessions,

you will fail the course!

Other four labs will be after MT exam.

Important!

- **At 6th Week:**
- **The lecture on 23rd March will be between 16 - 18:00.**
- **Quiz 1 will be at 5pm on 26th March!**
  - Quiz 1 Classrooms: Z11 ve 103

Basic Electric Circuits

Last Week

- Linearity And Superposition
- Thevenin and Norton

This Week

- Semiconductors
- Diodes (until midterm)

Basic Electric Circuits

Linearity and Superposition: Linearity

Basically, a mathematical equation is said to be linear if the following properties hold Circuits, Signals and Systems (CSS).

- **homogenity**
- **additivity**

What does this mean? We first look at the property of homogenity.

Basic Electric Circuits

Linearity : **Homogeneity.**

Homogeneity means that scaling the input of a system by a constant results in the output being scaled by the same constant.

Illustration: Does homogenity hold for the following equation?

Given,

$$y = 4x \qquad \text{Eq 3.1}$$

The equations that we think are linear in CSS, turn out not be linear because they fail the homogenity property. We next consider such an example.

Basic Electric Circuits

Linearity : **Homogeneity (scaling).**

Illustration: Does homogenity hold for the following equation?

Given,

$$y = 4x + 2 \qquad \text{Eq 3.2}$$

Basic Electric Circuits

Linearity : Homogeneity (scaling).

Illustration: Does homogenity hold for the following equation?

Given,

$$y = 4x + 2 \qquad \text{Eq 3.2}$$

If x = 1, then y = 6. If we double x to x=2, then y = 10.

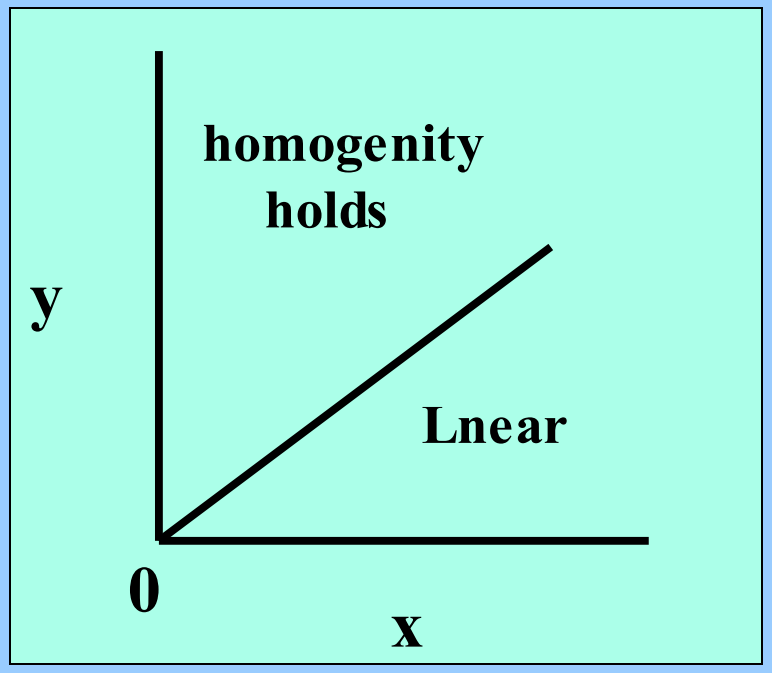
Now, since we doubled x we should be able to double the value that y had when x = 1 and get y = 10. In this case we get y = (2)(6) = 12, which obviously is not 10, so homogenity does not hold.

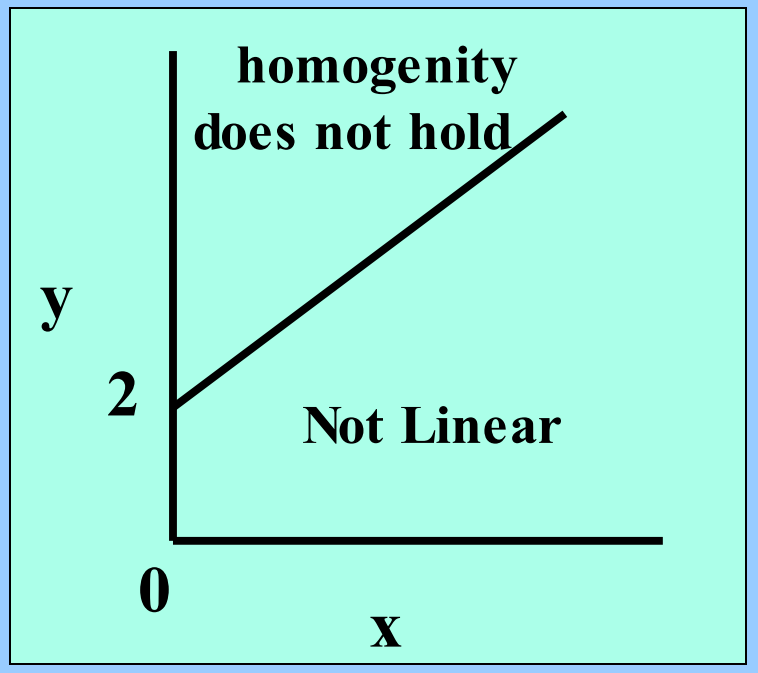
We conclude that Eq 3.2 is not a linear equation. In some ways that goes against the gain of what we have been taught about linear equations.

Basic Electric Circuits

Linearity : **Homogeneity (scaling).**

Many of us were brought-up to think that if plotting an equation yields a straight line, then the equation is linear. From the following illustrations we have;





Basic Electric Circuits

Linearity : **Additivity Property.**

Additivity means that the system's response to a sum of inputs equals the sum of its responses to each input applied separately.

This can be explained by considering the following illustrations.

Basic Electric Circuits

Linearity : Additivity Property.

Illustration: Given, $y = 4x$.

Let $x = x_1$, then $y_1 = 4x_1$
Let $x = x_2$, then $y_2 = 4x_2$

Then $y = y_1 + y_2 = 4x_1 + 4x_2$ Eq 3.3

Also, we note,

$$y = f(x_1 + x_2) = 4(x_1 + x_2) = 4x_1 + 4x_2 \quad \text{Eq 3.4}$$

Since Equations (3.3) and (3.4) are identical, the additivity property holds.

Basic Electric Circuits

Linearity : **Additivity Property.**

Illustration: **Given, $y = 4x + 2$.**

Is it additive?

Basic Electric Circuits

Linearity : Additivity Property.

Illustration: Given, $y = 4x + 2$.

Let $x = x_1$, then $y_1 = 4x_1 + 2$
Let $x = x_2$, then $y_2 = 4x_2 + 2$

Then $y = y_1 + y_2 = 4x_1+2 + 4x_2+2 = 4(x_1+x_2) + 4$ Eq 3.5

Also, we note,

$$y = f(x_1 + x_2) = 4(x_1 + x_2) + 2$$ Eq 3.6

Since Equations (3.5) and (3.6) are not identical, the additivity property does not hold.

Basic Electric Circuits

Superposition: Illustration using a circuit.

Consider the circuit below that contains two voltage sources.

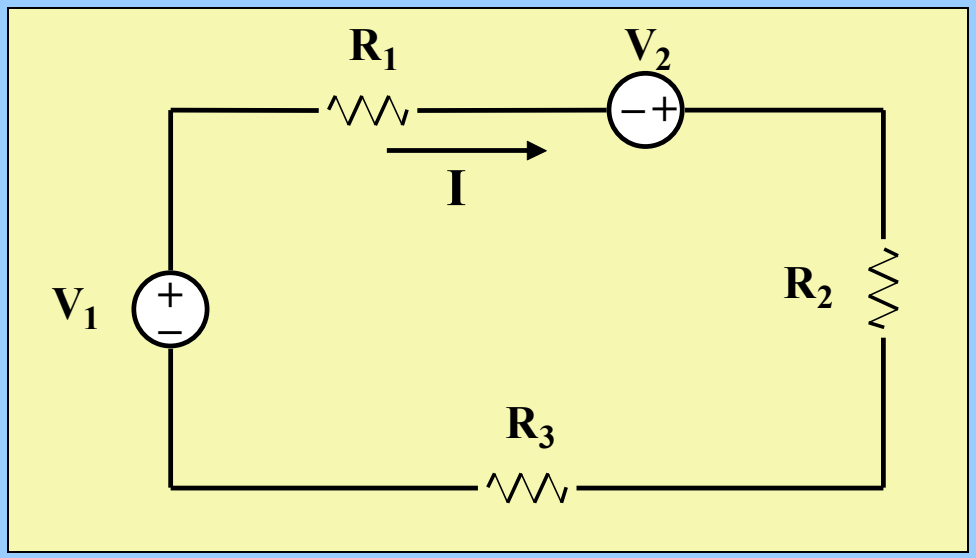
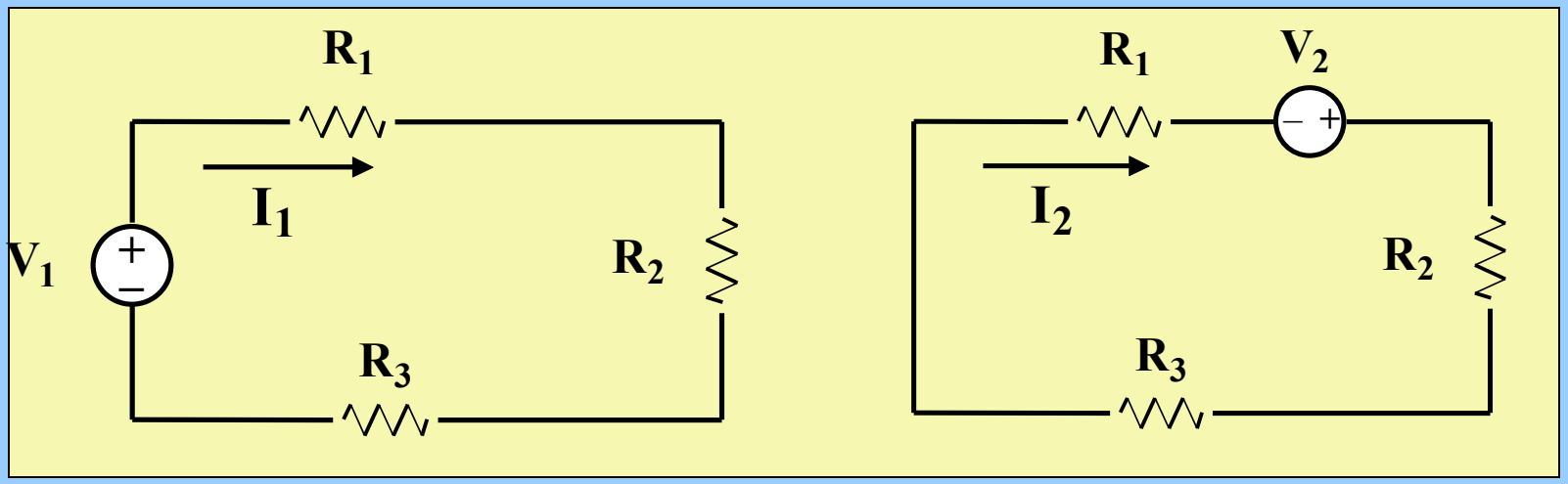


Figure 3.4: Circuit to illustrate superposition

We assume that V_1 and V_2 acting together produce current I.

Basic Electric Circuits

Superposition: Illustration using a circuit.



V_1 produces current I_1

V_2 produces current I_2

Superposition states that the current, I, produced by both sources acting together (Fig 3.4) is the same as the sum of the currents, $I_1 + I_2$, where I_1 is produced by V_1 and I_2 is produced by V_2.

THEVENIN & NORTON

THEVENIN'S THEOREM: Example 10.1.

Find V_X by first finding V_{TH} and R_{TH} to the left of A-B.

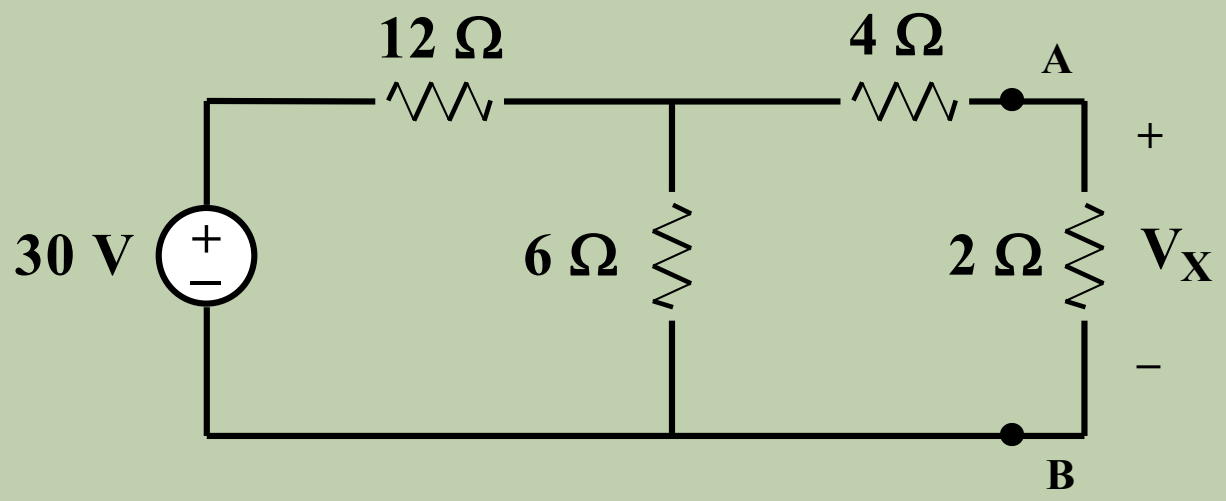


Figure 10.8: Circuit for Example 10.1.

First remove everything to the right of A-B.

THEVENIN & NORTON

THEVENIN'S THEOREM: Example 10.1. continued

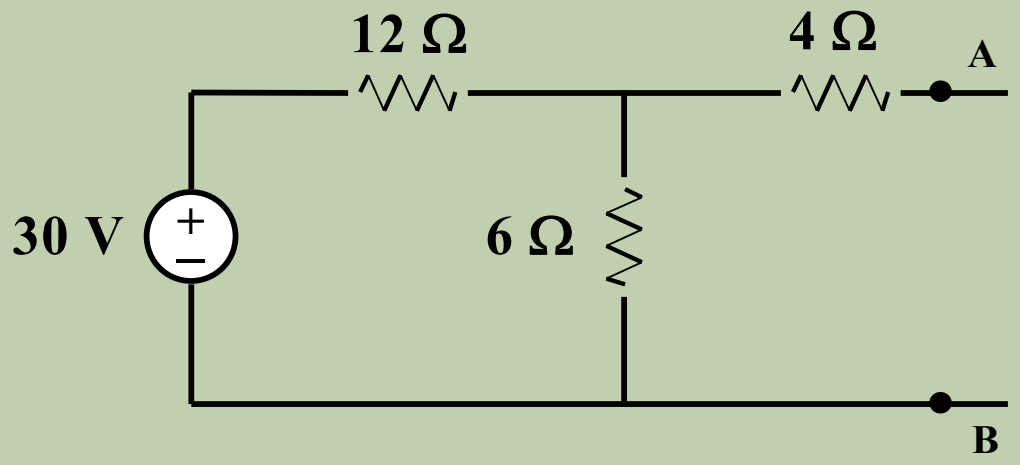


Figure 10.9: Circuit for finding V_{TH} for Example 10.1.

$$V_{AB} = \frac{(30)(6)}{6+12} = 10V$$

Notice that there is no current flowing in the 4 Ω resistor (A-B) is open. Thus there can be no voltage across the resistor.

THEVENIN & NORTON

THEVENIN'S THEOREM: Example 10.1. continued

We now deactivate the sources to the left of A-B and find the resistance seen looking in these terminals.

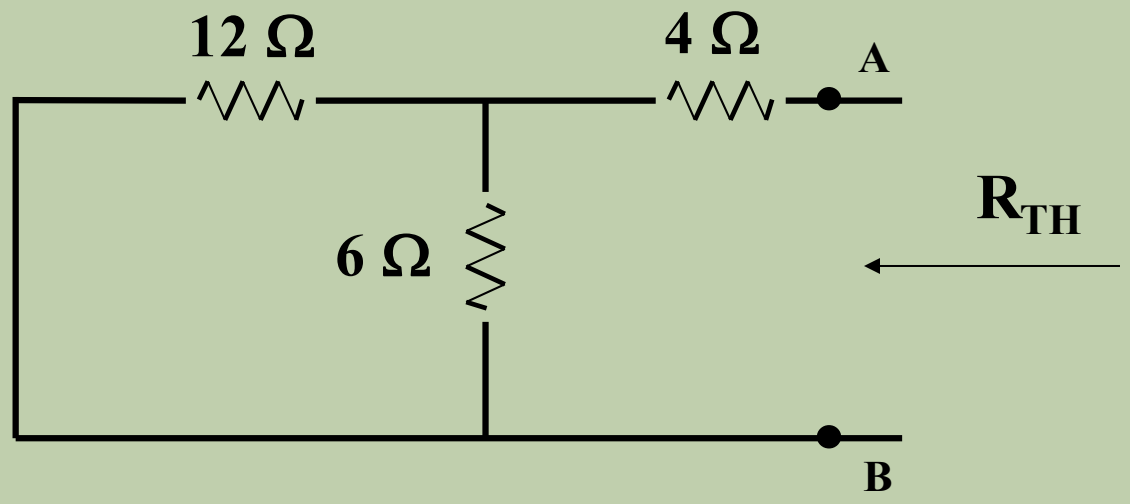


Figure 10.10: Circuit for find R_{TH} for Example 10.10.

We see,

$$R_{TH} = 12||6 + 4 = 8\,\Omega$$

THEVENIN & NORTON

THEVENIN'S THEOREM: Example 10.1. continued

After having found the Thevenin circuit, we connect this to the load in order to find V_X.

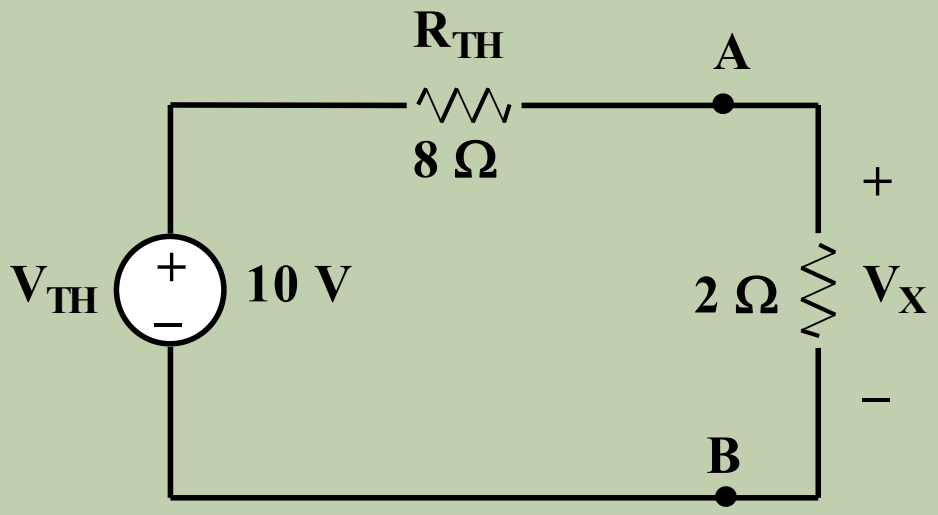


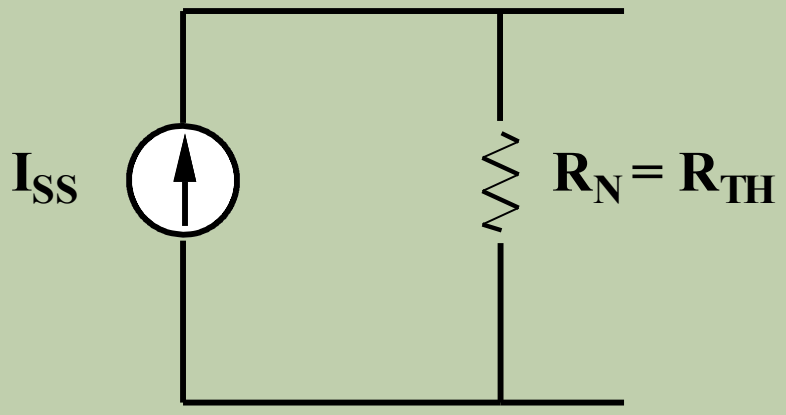
Figure 10.11: Circuit of Ex 10.1 after connecting Thevenin circuit.

$$V_X = \frac{(10)(2)}{2+8} = 2V$$

THEVENIN & NORTON

NORTON'S THEOREM:

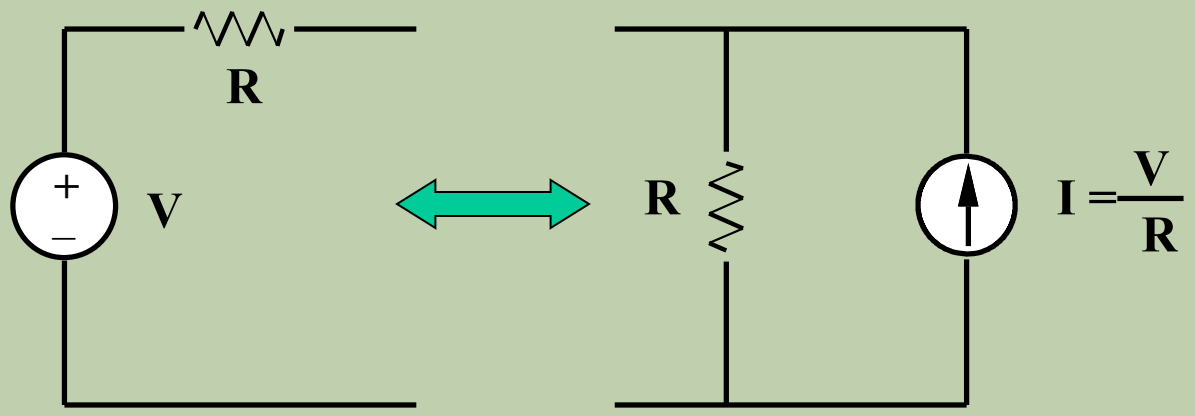
In the Norton circuit, the current source is the short circuit current of the network, that is, the current obtained by shorting the output of the network. The resistance is the resistance seen looking into the network with all sources deactivated. This is the same as R_{TH}.



THEVENIN & NORTON

NORTON'S THEOREM:

We recall the following from source transformations.



In view of the above, if we have the Thevenin equivalent circuit of a network, we can obtain the Norton equivalent by using source transformation.

However, this is not how we normally go about finding the Norton equivalent circuit.

THEVENIN & NORTON

NORTON'S THEOREM: Example 10.6.

Find the Norton equivalent circuit to the left of terminals A-B for the network shown below. Connect the Norton equivalent circuit to the load and find the current in the 50 Ω resistor.

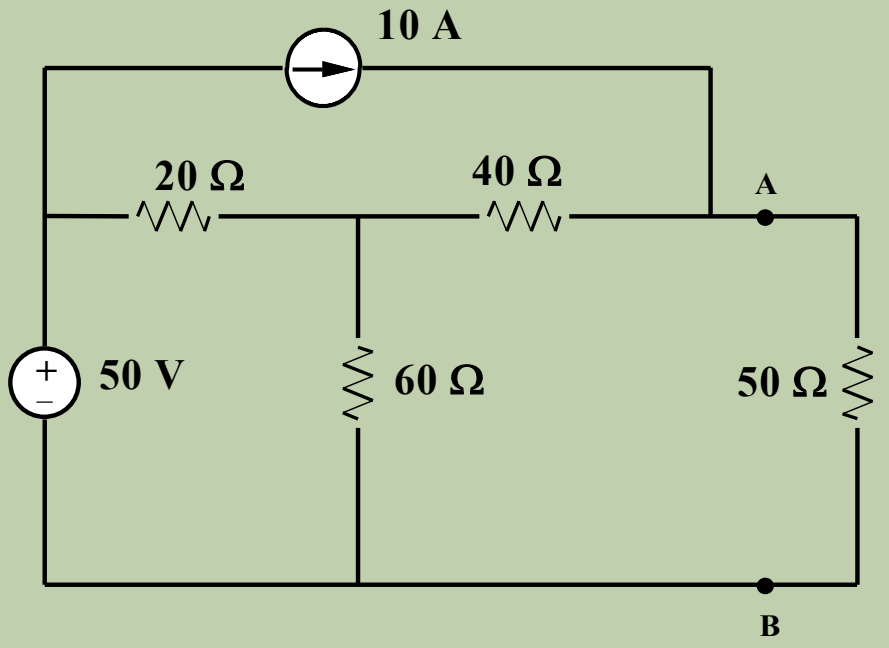


Figure 10.30: Circuit for Example 10.6.

THEVENIN & NORTON

NORTON'S THEOREM: Example 10.6. continued

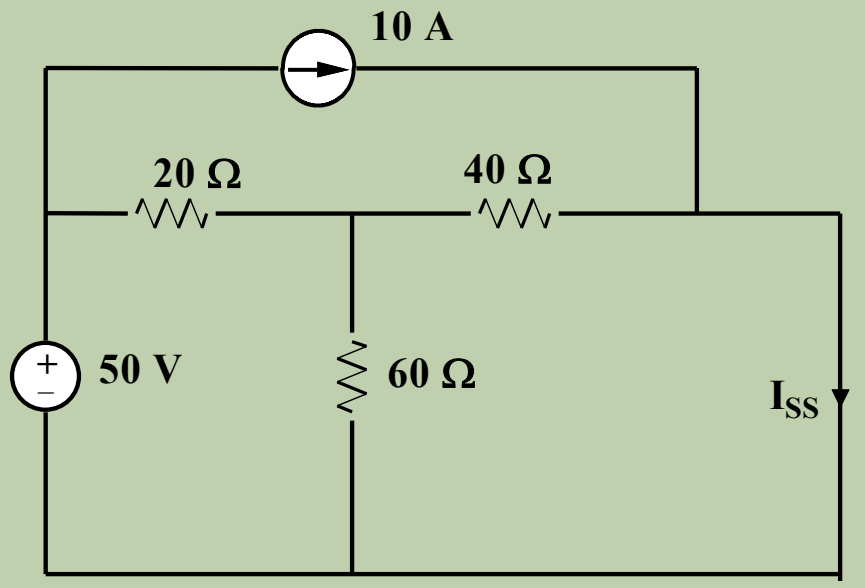


Figure 10.31: Circuit for find I_{NORTON}.

It can be shown by standard circuit analysis that

$$I_{SS}=10.7\ A$$

THEVENIN & NORTON

NORTON'S THEOREM: Example 10.6. continued

It can also be shown that by deactivating the sources, We find the resistance looking into terminals A-B is

$$R_N = 55\ \Omega$$

R_N and R_{TH} will always be the same value for a given circuit. The Norton equivalent circuit tied to the load is shown below.

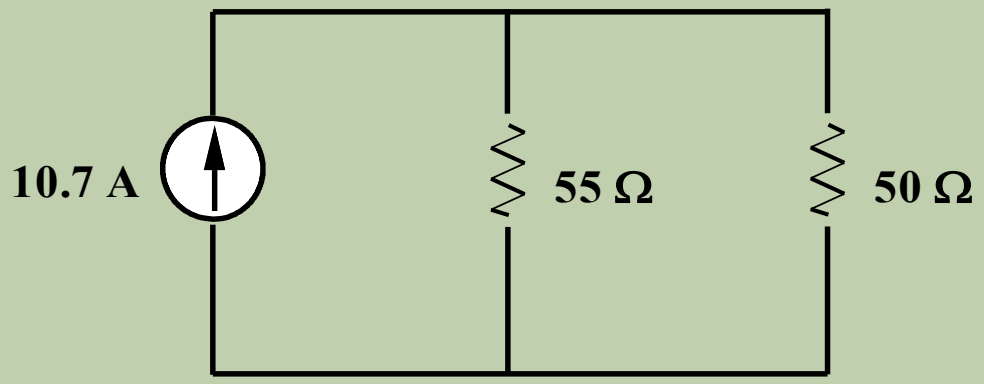


Figure 10.32: Final circuit for Example 10.6.

Basic Electric Circuits

Linearity :

Informally,
a linear circuit is one in which the electronic components' values (such as resistance, capacitance, inductance, gain, etc.) do not change with the level of voltage or current in the circuit.
A circuit is linear if it consists only of:

- **Resistors (R)**
- **Inductors (L)**
- **Capacitors (C)**
- **Independent and dependent voltage/current sources (provided they follow linear relationships).**

- **We can apply superposition and Thevenin Norton Theorems if the circuit is linear.**

Non-linear Circuits

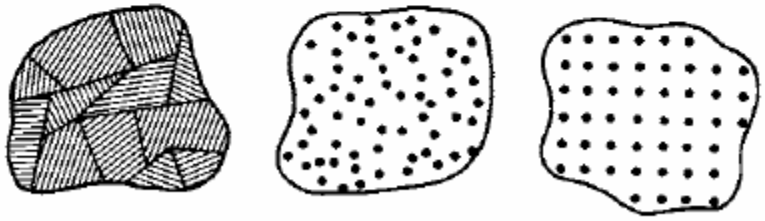
Mainly comprised of:

- **Diodes**
- **Transistors**

They are made with semiconductor materials.

What is a Semiconductor?

- Low resistivity => “conductor”
- High resistivity => “insulator”
- Intermediate resistivity => “semiconductor”
  - conductivity lies between that of conductors and insulators
  - generally crystalline in structure for IC devices
    - In recent years, however, non-crystalline semiconductors have become commercially very important



polycrystalline amorphous crystalline

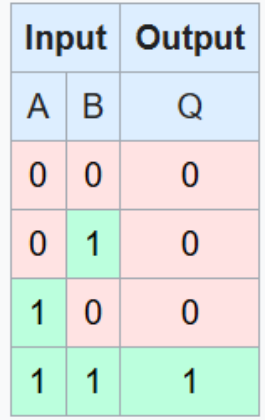
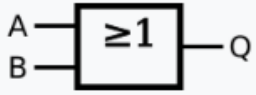
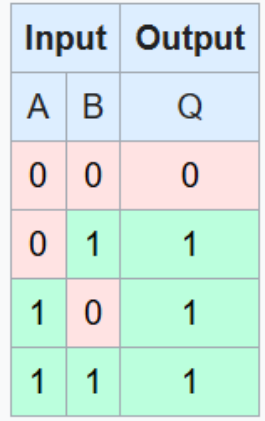
Semiconductors

It is a revolution!

Why?

Semiconductors

We can make Logic Gates with diodes and transistors.

AND			$A \cdot B$ or $A \wedge B$	
OR			$A + B$ or $A \vee B$	

Input		Output
A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Input		Output
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

Semiconductors

CPUs consist of Aritmetic Logic Units – made up with logic gates.

Our software runs on the processors which are made up with many transistors.

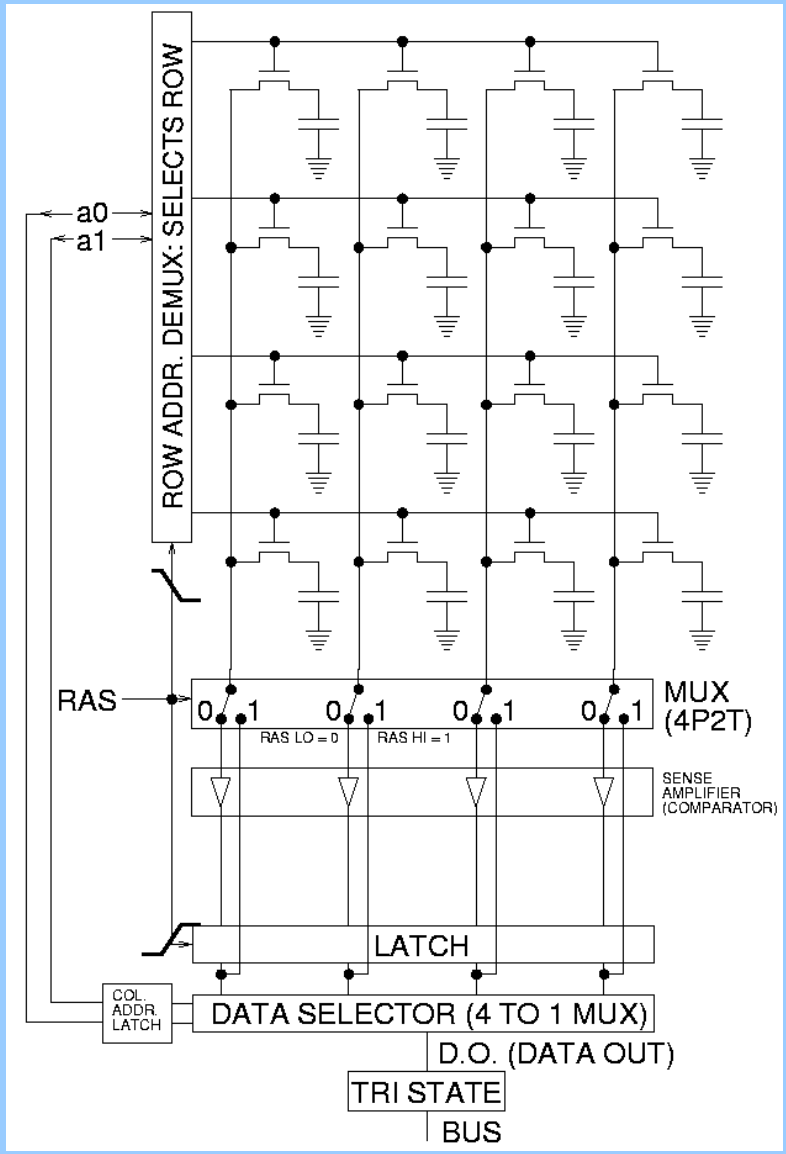
Intel I7 processor has 731 million transistors on it.



FIG. 1.2
Intel® Core™ i7 Extreme Edition Processor.

Semiconductors

Memories are made up with transistors and logic gates.

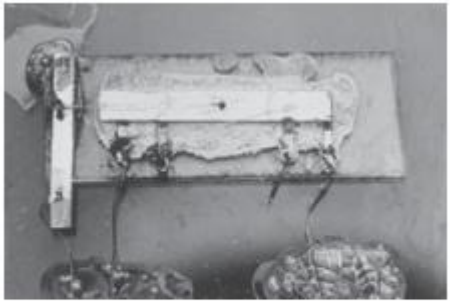


Semiconductors

Our current modern PCs could be made possible thanks to semiconductors.



Semiconductors



The first integrated circuit, a phase-shift oscillator, invented by Jack S. Kilby in 1958. (Courtesy of Texas Instruments.)



Jack St. Clair Kilby, inventor of the integrated circuit and co-inventor of the electronic handheld calculator. (Courtesy of Texas Instruments.)

Born: Jefferson City, Missouri, 1923. MS, University of Wisconsin. Director of Engineering and Technology, Components Group, Texas Instruments. Fellow of the IEEE. Holds more than 60 U.S. patents.

Semiconductors

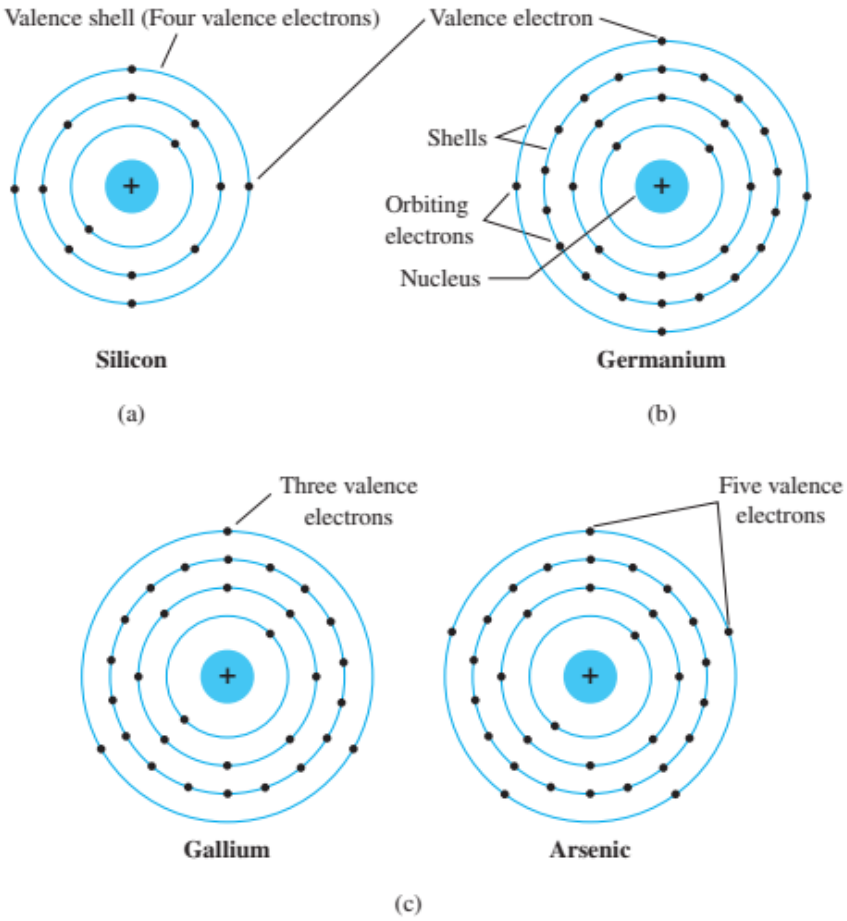


FIG. 1.3
Atomic structure of (a) silicon; (b) germanium; and (c) gallium and arsenic.

Semiconductors

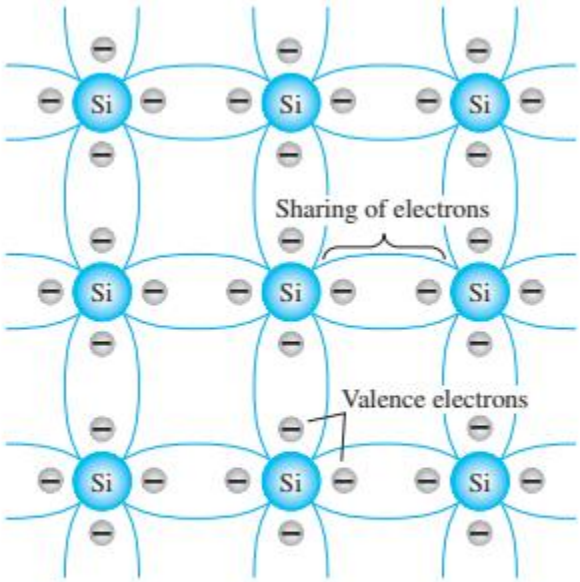


FIG. 1.4

Covalent bonding of the silicon atom.

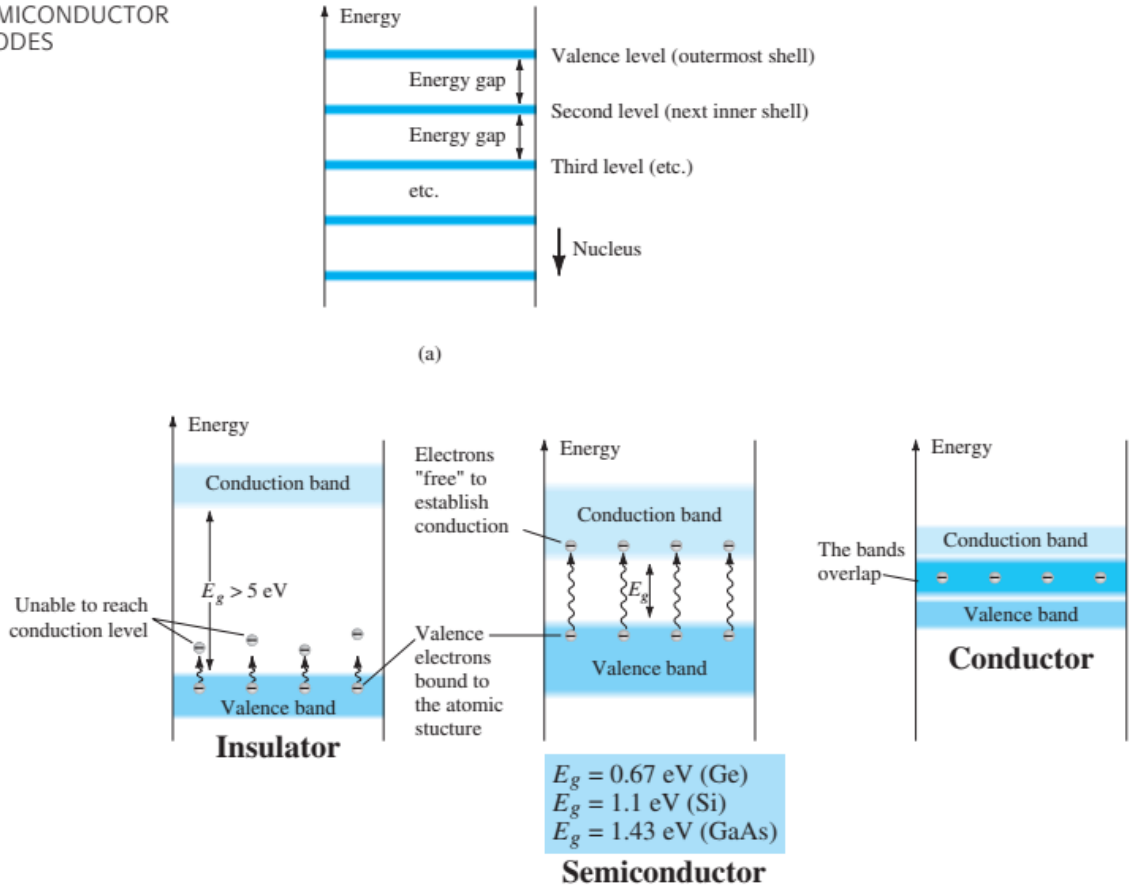


FIG. 1.6

Energy levels: (a) discrete levels in isolated atomic structures; (b) conduction and valence bands of an insulator, a semiconductor, and a conductor.

TABLE 1.1

Intrinsic Carriers n_i

Semiconductor	Intrinsic Carriers (per cubic centimeter)
GaAs	1.7×10^6
Si	1.5×10^{10}
Ge	2.5×10^{13}

TABLE 1.2

Relative Mobility Factor μ_n

Semiconductor	μ_n ($cm^2/V \cdot s$)
Si	1500
Ge	3900
GaAs	8500

n-type material

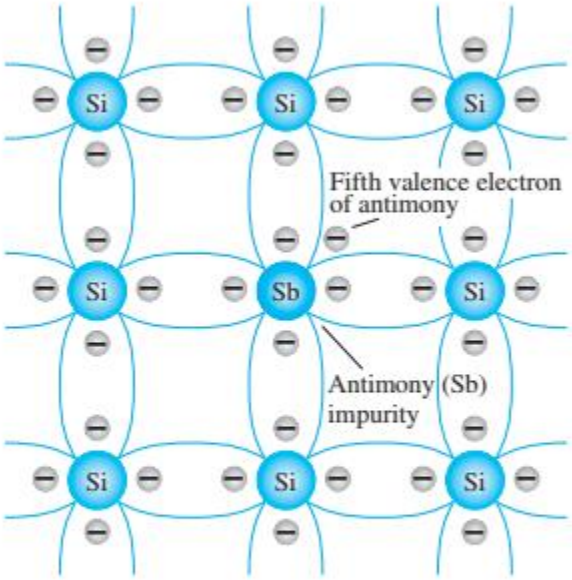


FIG. 1.7

Antimony impurity in n-type material.

n-type material

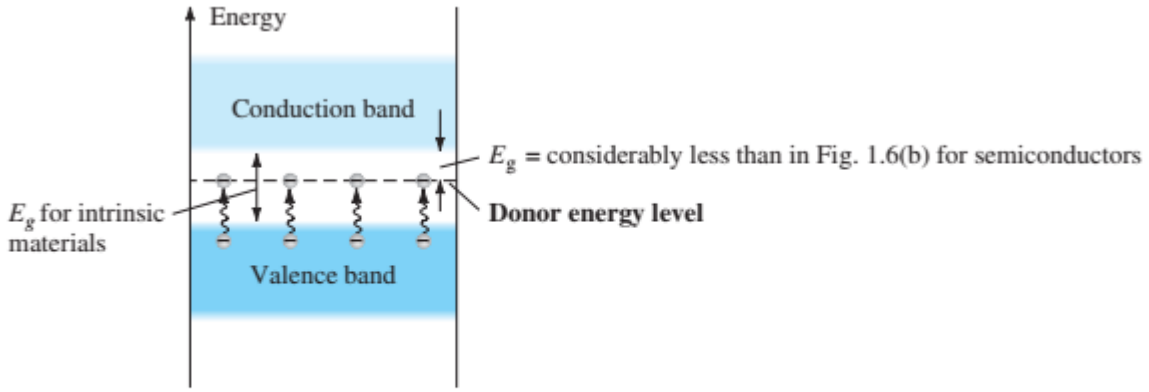


FIG. 1.8
Effect of donor impurities on the energy band structure.

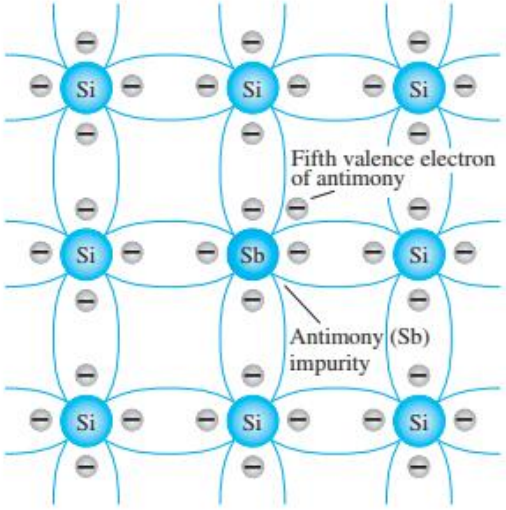


FIG. 1.7
Antimony impurity in n-type material.

p-type material

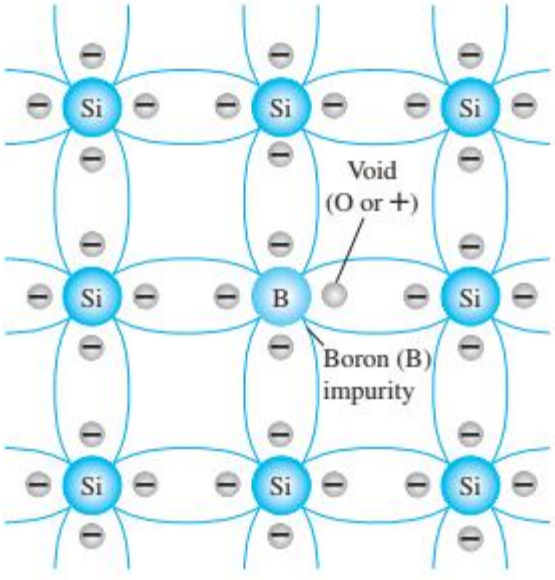


FIG. 1.9
Boron impurity in p-type material.

p-type material

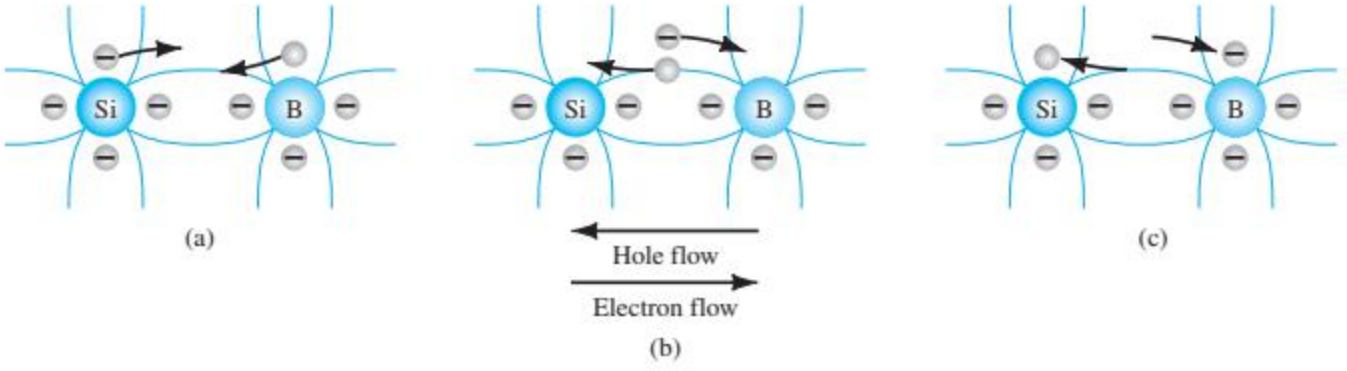
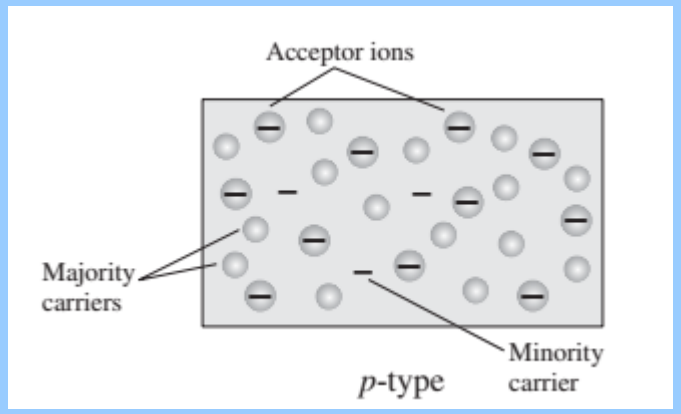
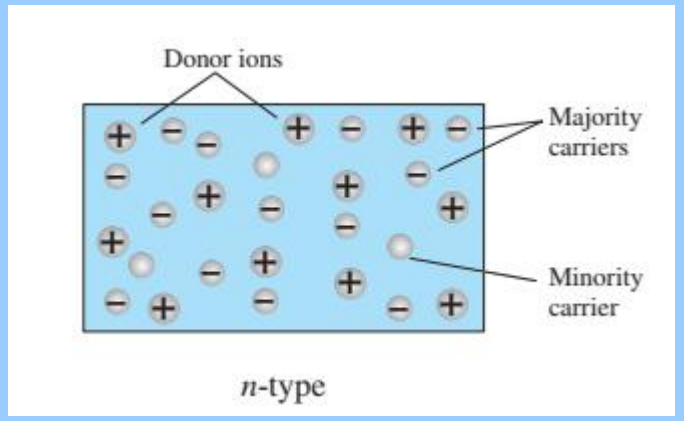


FIG. 1.10
Electron versus hole flow.

Semiconductors





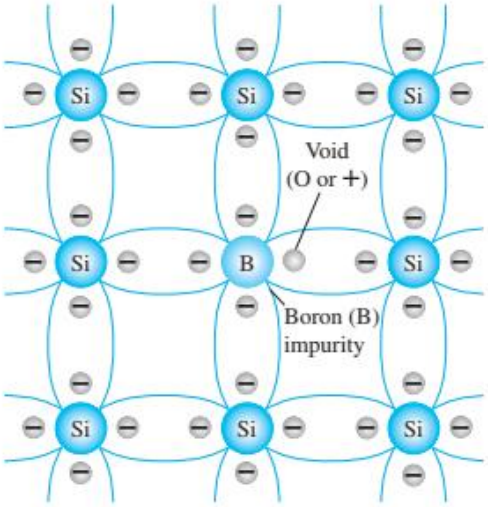


FIG. 1.9
Boron impurity in p-type material.

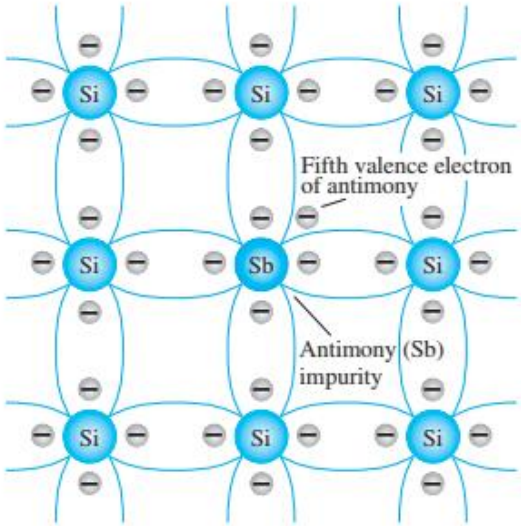
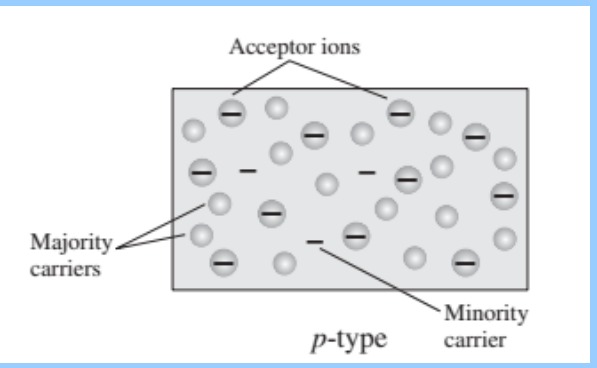
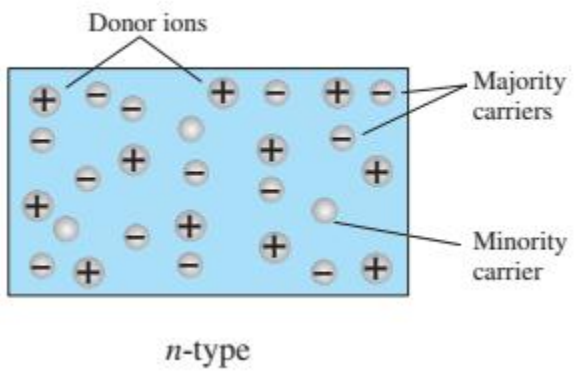


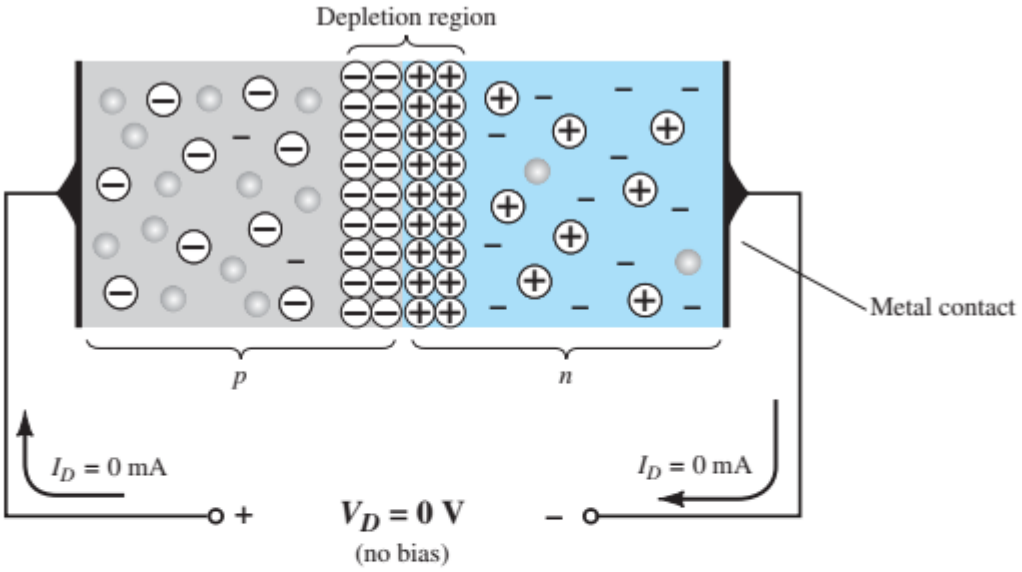
FIG. 1.7
Antimony impurity in n-type material.

Diode

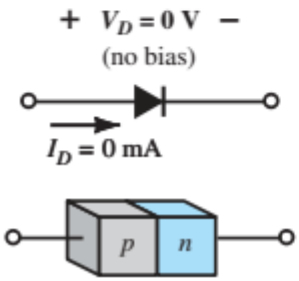
$V_D = 0V$

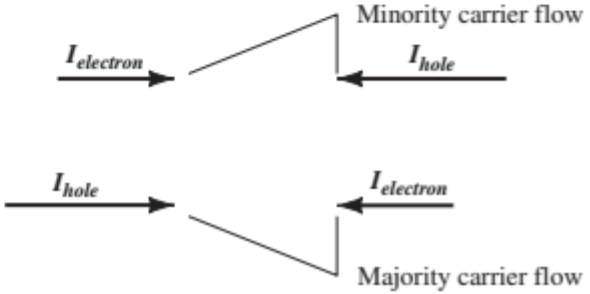

Acceptor ions
Majority carriers
Minority carrier
p-type


Donor ions
Majority carriers
Minority carrier
n-type


Depletion region
Metal contact
p
n
$I_D = 0$ mA
$I_D = 0$ mA
+
$V_D = 0$ V
(no bias)
–

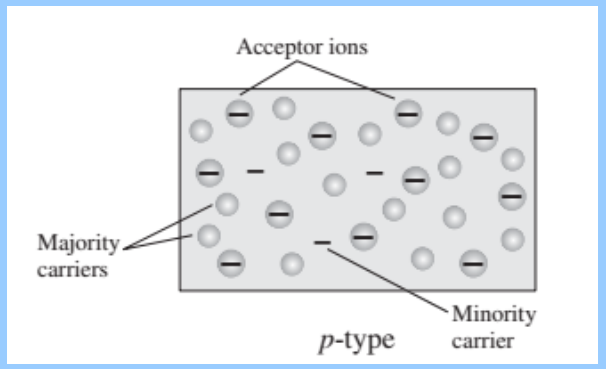
(a)

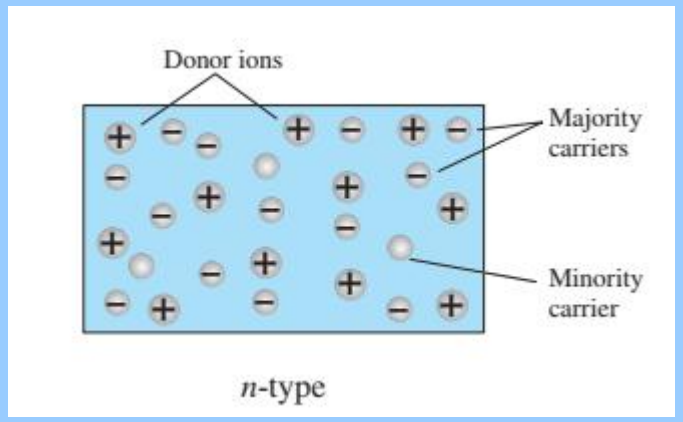

+ $V_D = 0$ V –
(no bias)
$I_D = 0$ mA
p
n

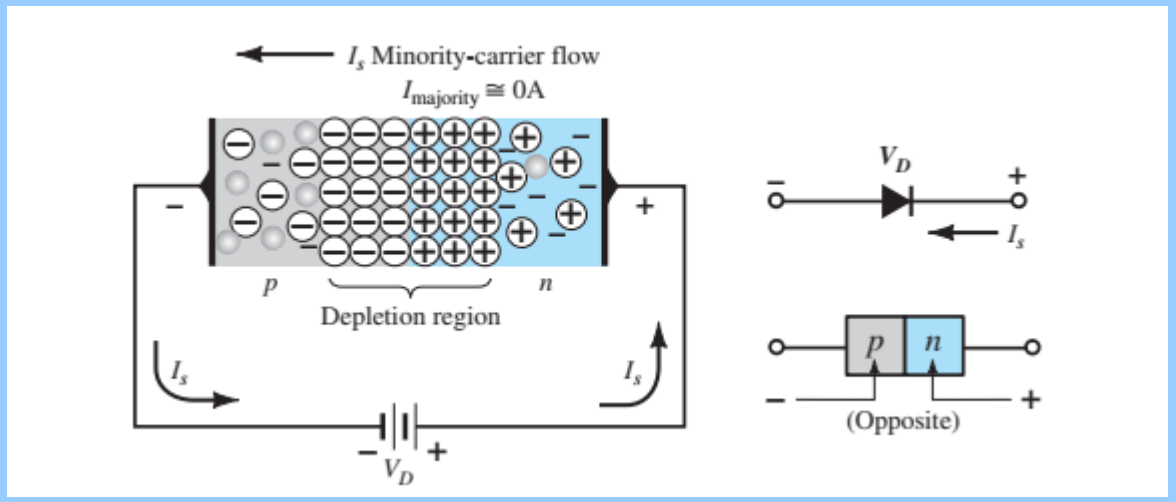

Minority carrier flow
$I_{electron}$
I_{hole}
I_{hole}
$I_{electron}$
Majority carrier flow

Diode

$V_D < 0V$

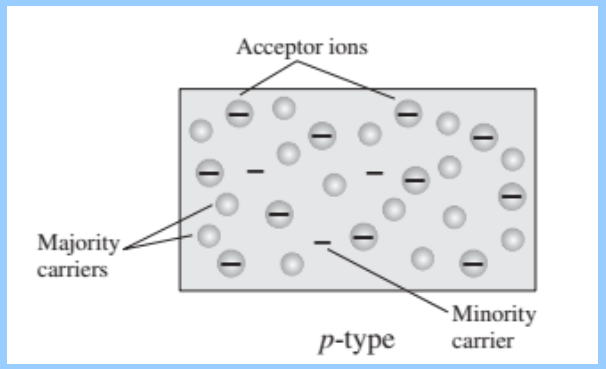

Acceptor ions
Majority carriers
Minority carrier
p-type

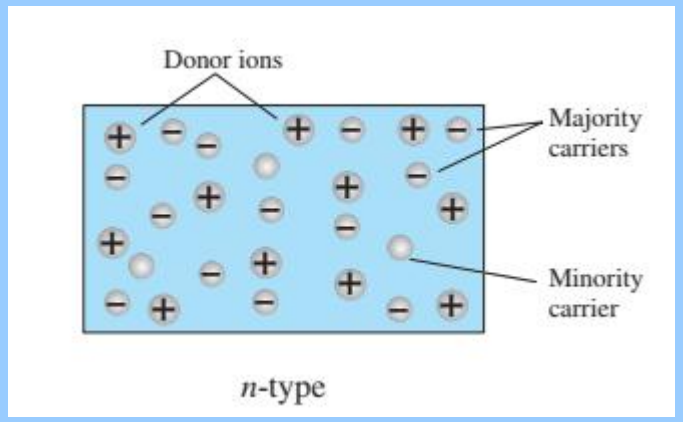

Donor ions
Majority carriers
Minority carrier
n-type

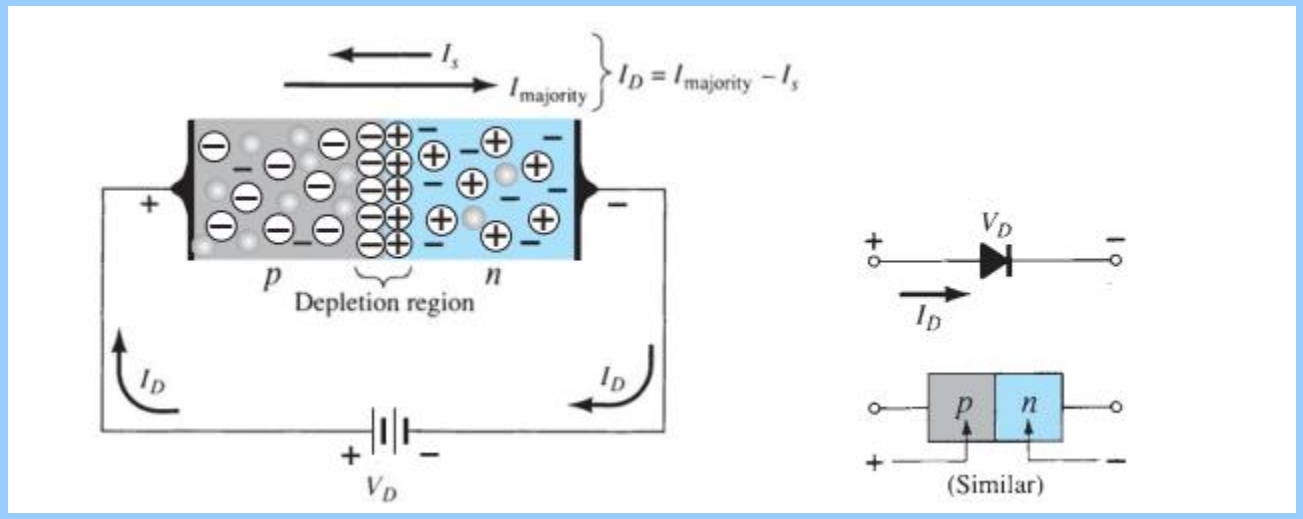

I_s Minority-carrier flow
$I_{majority} \cong 0A$
p
n
Depletion region
I_s
I_s
V_D
V_D
I_s
p
n
(Opposite)

Diode

$V_D > 0V$


Acceptor ions
Majority carriers
Minority carrier
p-type


Donor ions
Majority carriers
Minority carrier
n-type


I_s
$I_{majority}$
$I_D = I_{majority} - I_s$
p
n
Depletion region
I_D
I_D
V_D
V_D
I_D
p
n
(Similar)

Diode

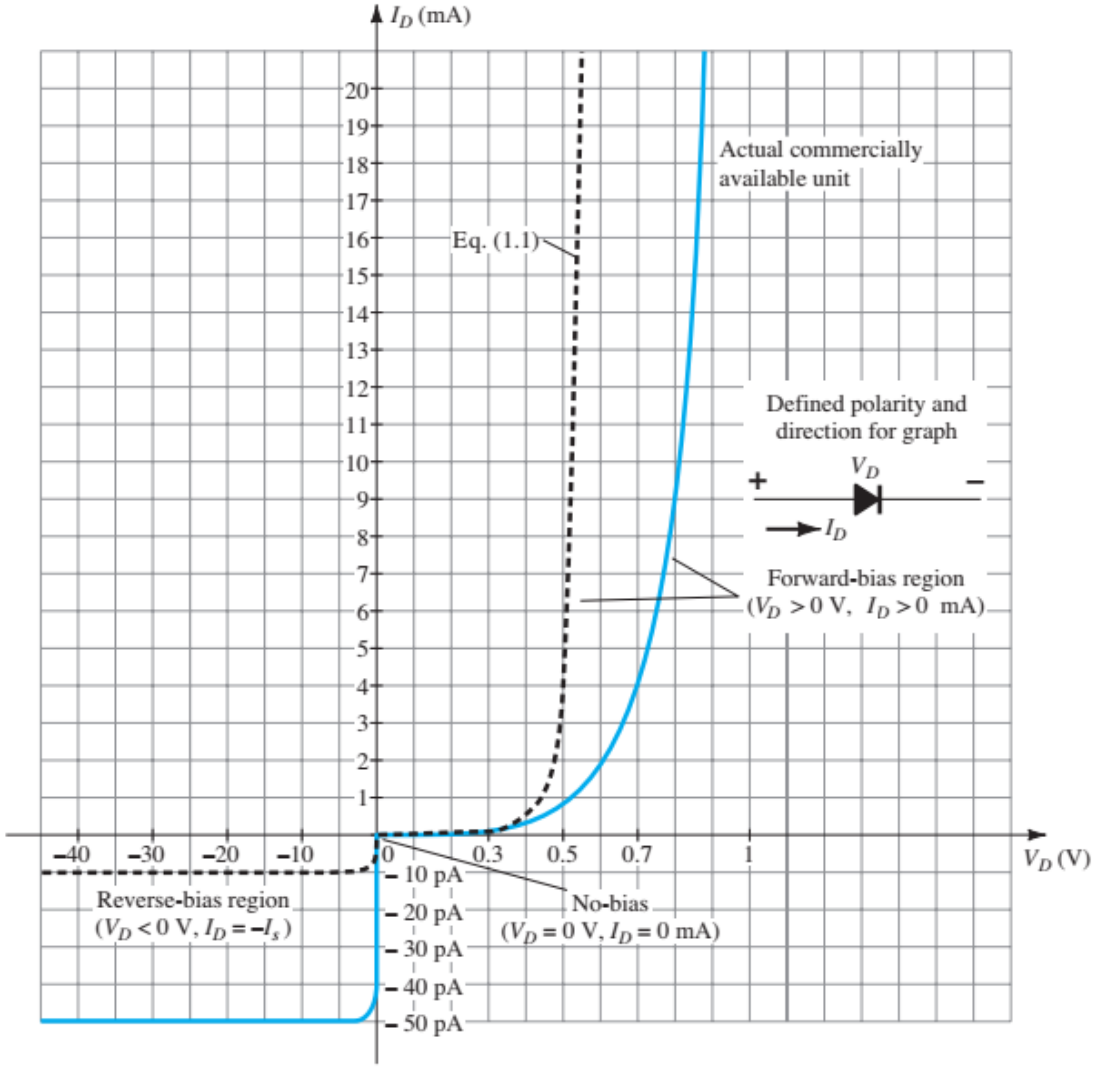


FIG. 1.15

Silicon semiconductor diode characteristics.

Diode

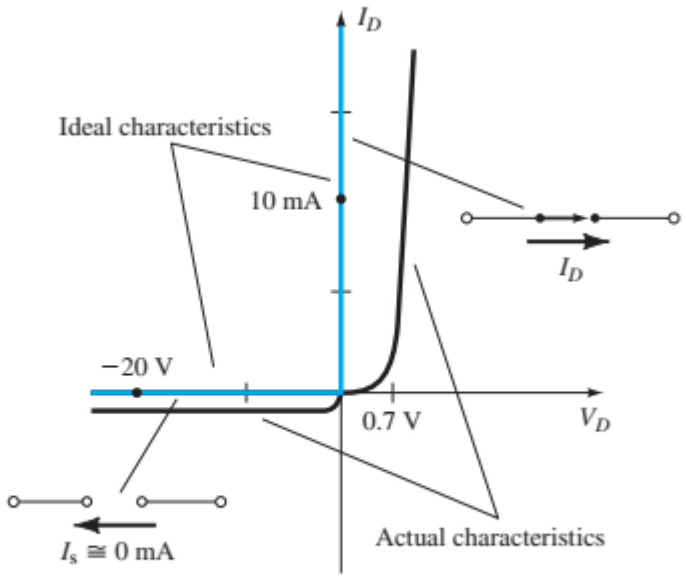


FIG. 1.22
Ideal versus actual semiconductor characteristics.

Diode

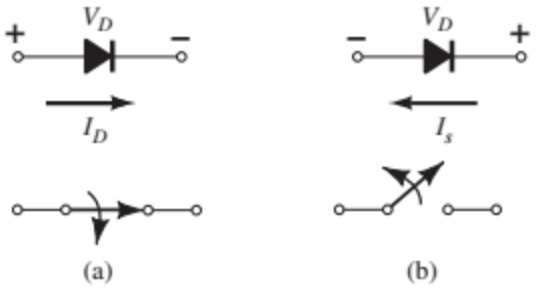
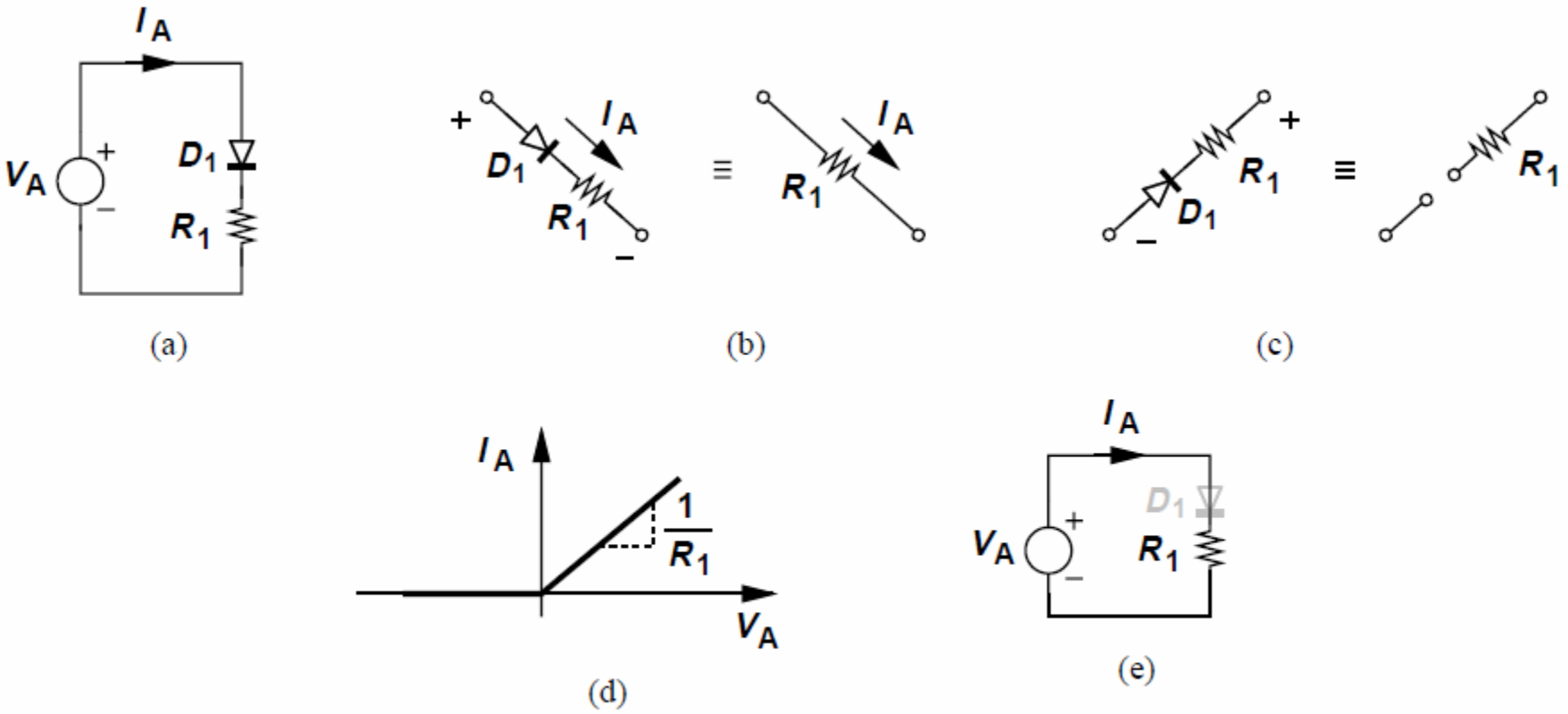


FIG. 1.21

Ideal semiconductor diode: (a) forward-biased; (b) reverse-biased.

EXAMPLE

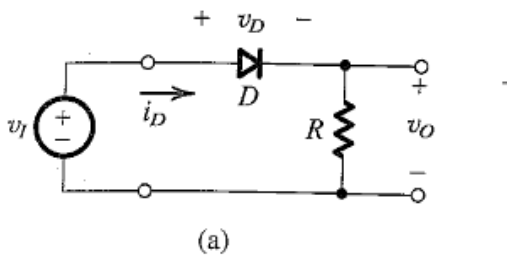
- Plot the I/V characteristic for the diode-resistor combination.



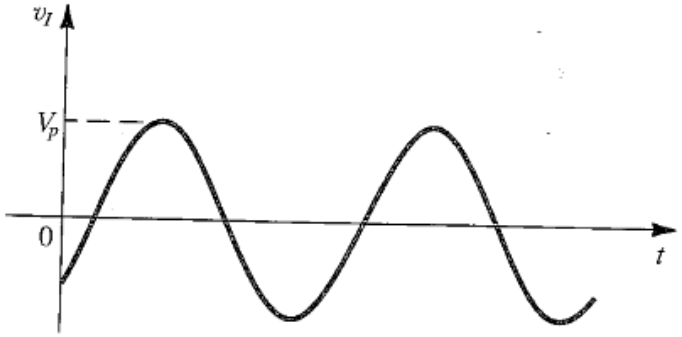
(a) Diode-resistor series combination, (b) equivalent circuit under forward bias, (c) equivalent circuit under reverse bias, (d) I/V characteristic, (e) equivalent circuit if D_1 is on.

EXAMPLE

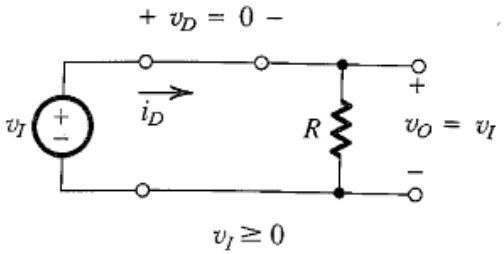
- Plot the I/V characteristic for the diode-resistor combination.



(a)

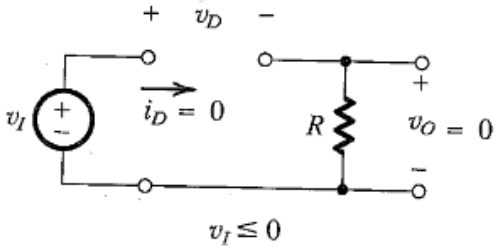


(b)



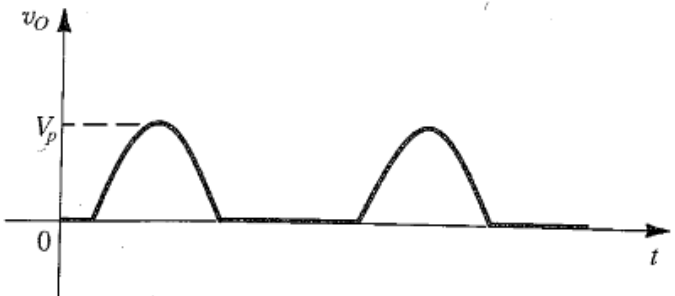
$v_I \geq 0$

(c)

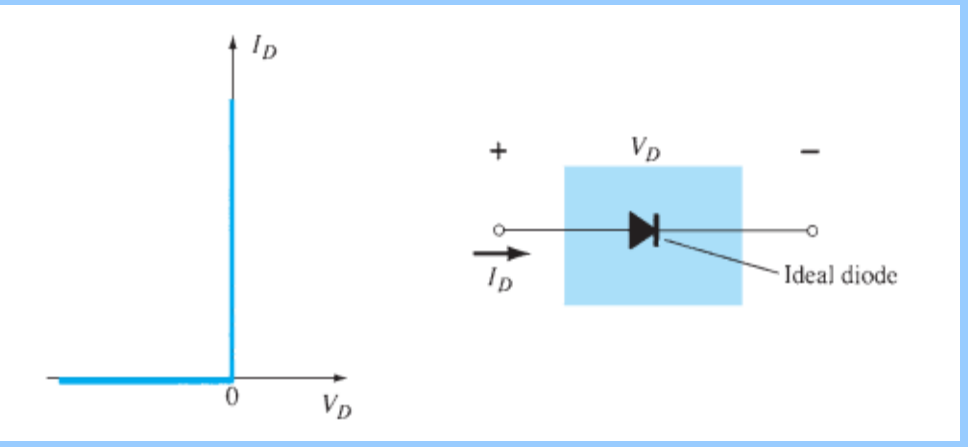


$v_I \leq 0$

(d)



Diode



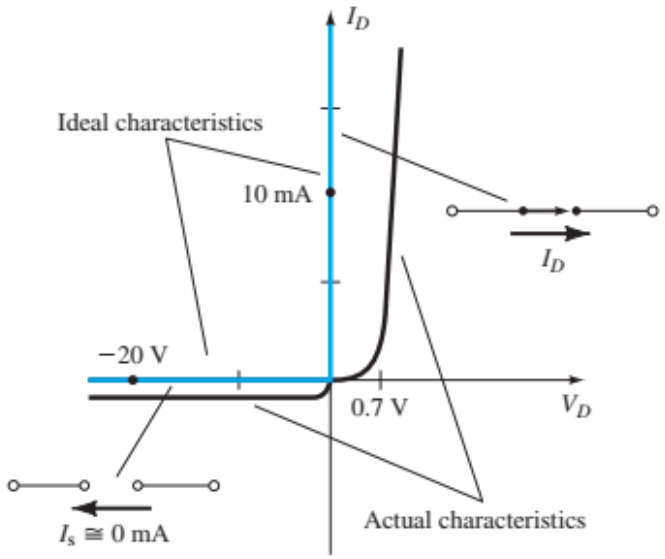
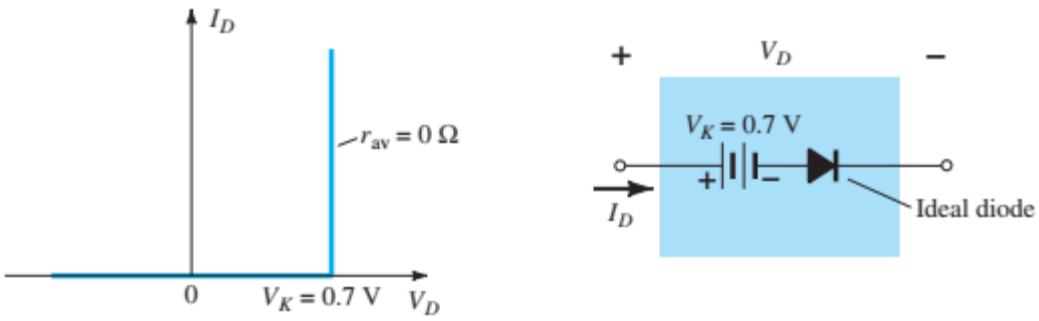


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode



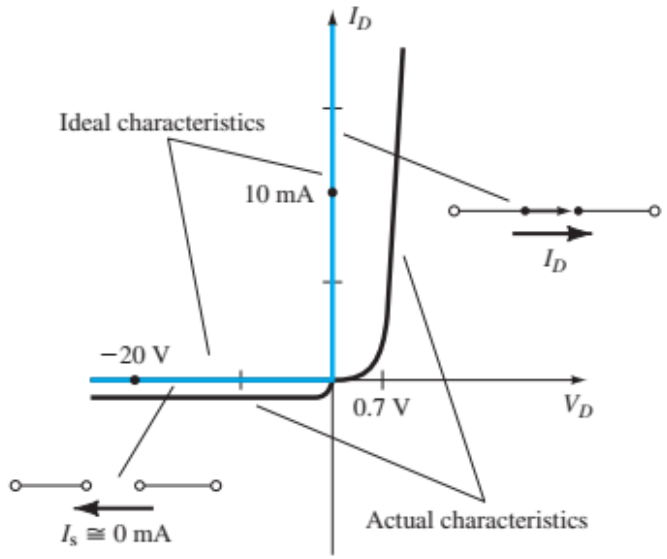


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode

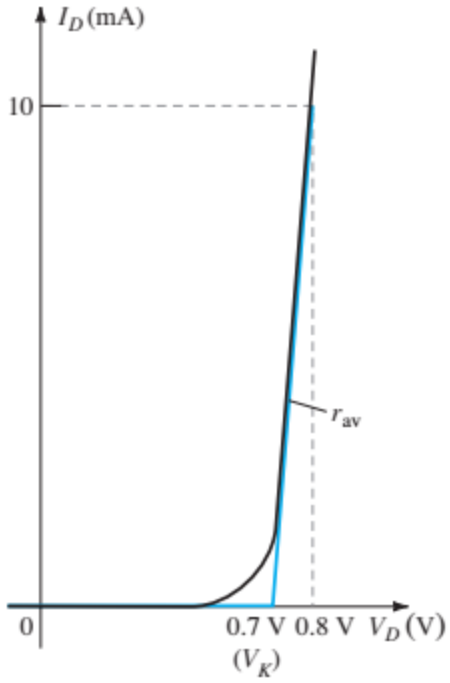
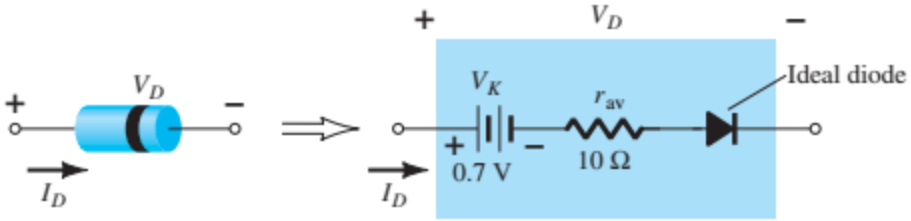


FIG. 1.29

Defining the piecewise-linear equivalent circuit using straight-line segments to approximate the characteristic curve.



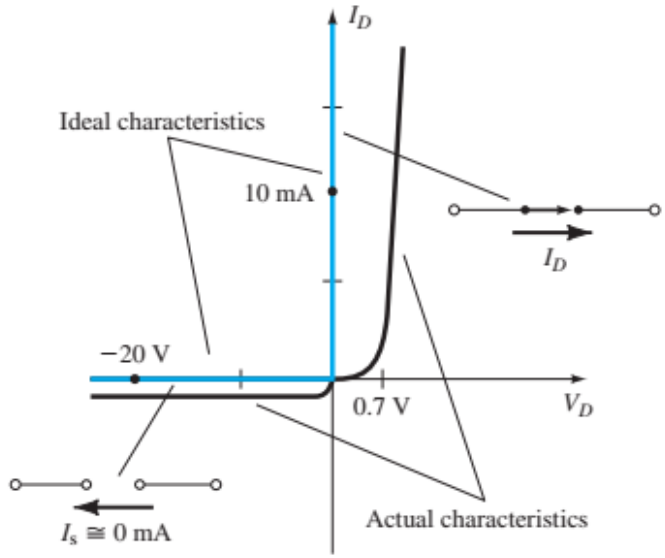
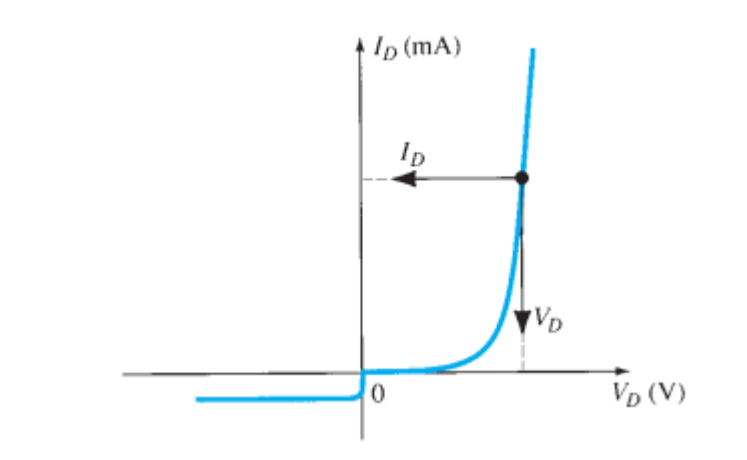


FIG. 1.22

Ideal versus actual semiconductor characteristics.

Diode – DC resistance

$$R_D = \frac{V_D}{I_D}$$



Diode – DC resistance

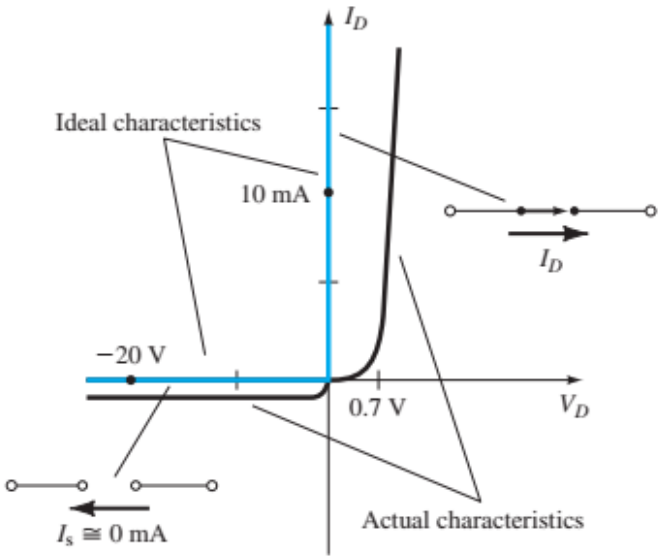
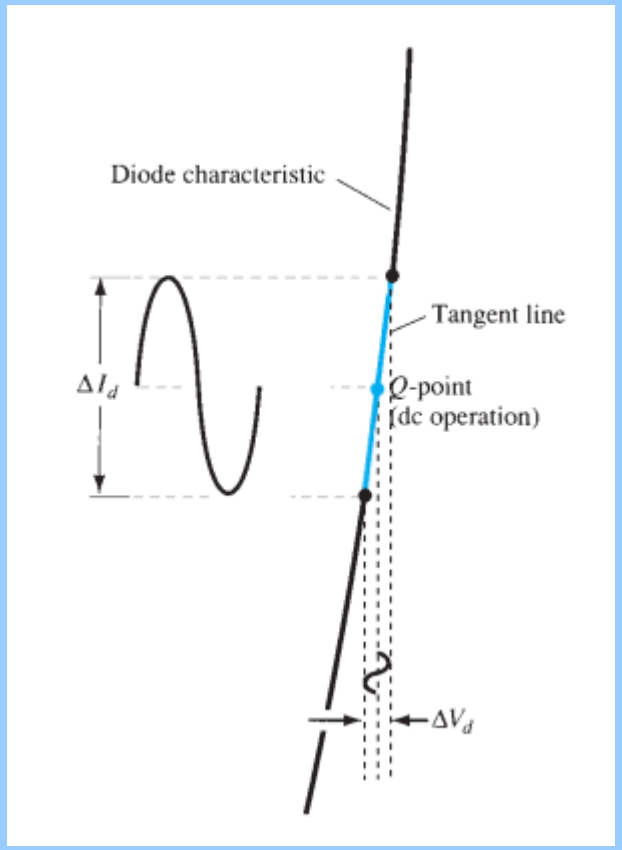
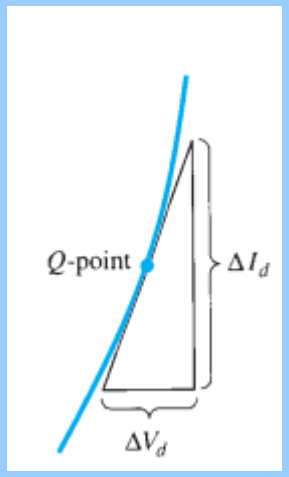


FIG. 1.22
Ideal versus actual semiconductor characteristics.



$$r_d = \frac{\Delta V_d}{\Delta I_d}$$



Diode – AC resistance

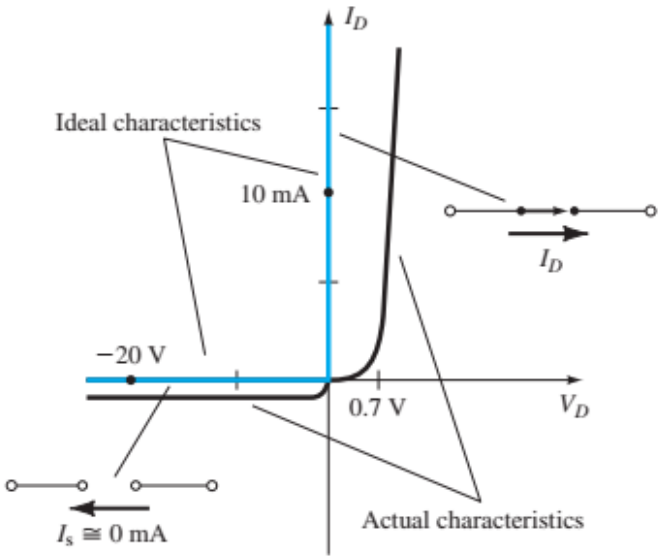
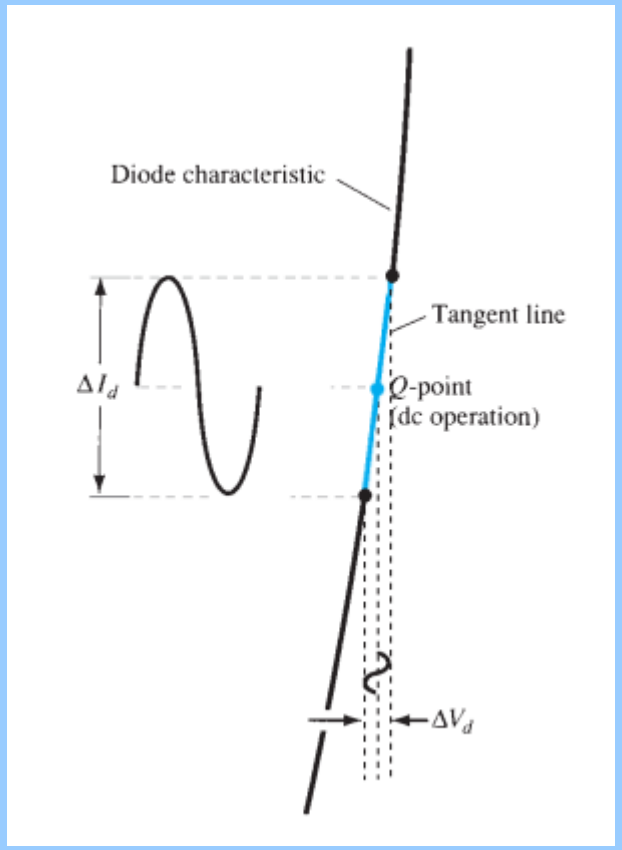
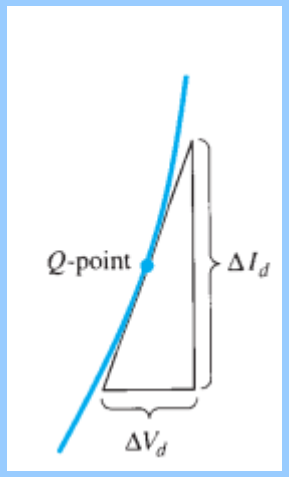


FIG. 1.22
Ideal versus actual semiconductor characteristics.



$$r_d = \frac{\Delta V_d}{\Delta I_d}$$



Diode – Average AC resistance

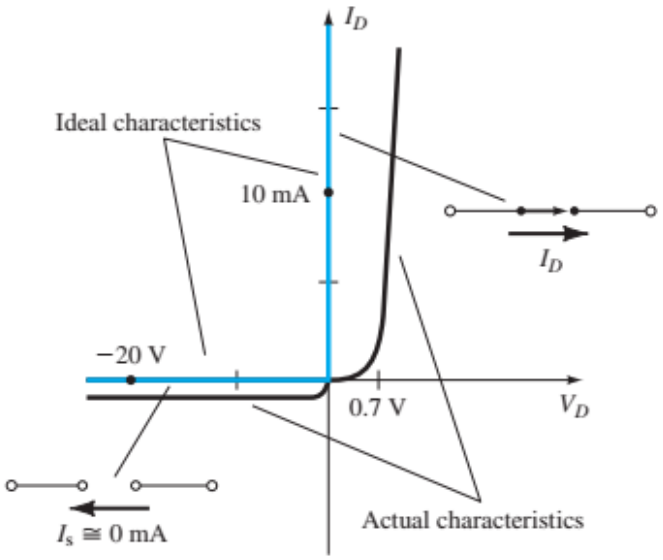
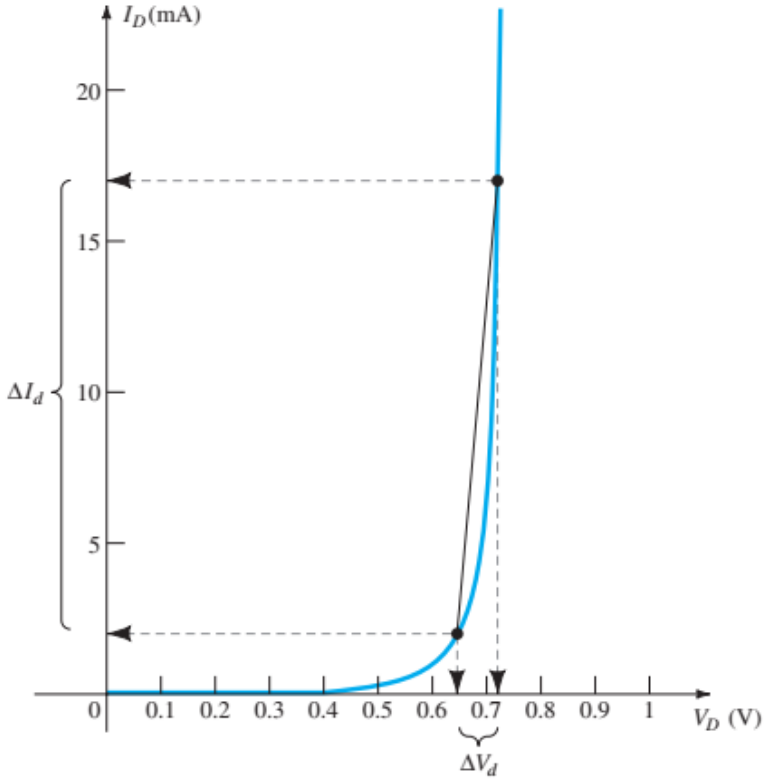
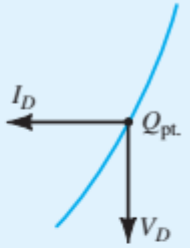
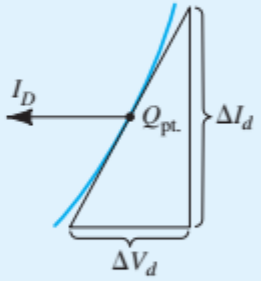
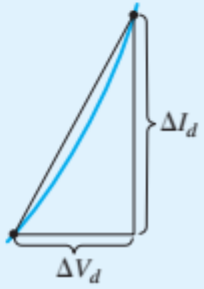


FIG. 1.22
Ideal versus actual semiconductor characteristics.



$$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right|_{\text{pt. to pt.}}$$

Diode – Resistance Levels

Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26\text{ mV}}{I_D}$	Defined by a tangent line at the Q-point	
Average ac	$r_{av} = \left.\frac{\Delta V_d}{\Delta I_d}\right\|_{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

Diode – Resistance Levels

